

# Junjie Zhao

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## EDUCATION

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### University of Rochester

*Bachelor of Science, Computer Science*

- Current GPA: 3.99 out of 4.00

**Rochester, NY**

*Anticipated May 2027*

## HONORS AND AWARDS

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- Dean's List: Fall 2023, Spring 2024, Fall 2024, Spring 2025, Fall 2025, Spring 2026
- Honorable Mentions for the 2026 CRA Outstanding Undergraduate Researcher Award
- Phi Beta Kappa, Elected in Junior Year

## RESEARCH EXPERIENCE

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### Department of Computer Science, University of Rochester

*Undergraduate Research Assistant in Horizon Lab*

*Supervised by Prof. Yuhao Zhu*

**Rochester, NY**

*Feb 2025 – Present*

- Lead the project on an open, wearable, end-to-end wireless varifocal AR research platform for XR systems and human-centered computing research.
- Designed and prototyped the AR hardware using Fusion 360 and 3D printing, integrating a Linux-based client, IMU, outward-facing camera, eye-tracking camera, and varifocal display.
- Built a client-server sensing/rendering pipeline for wireless camera and IMU streaming, server-side localization/rendering, and frame transmission back to the wearable display.
- Conducted Zemax simulations of virtual image formation, including FOV, distortion, and third-order aberration analysis, to guide optical and system-level design trade-offs.

*Undergraduate Research Assistant in ROC-HCI Lab*

*Supervised by Dr. Masum Hasan and Prof. Ehsan Hoque*

*Sep 2024 – Present*

- Contributed to HAL, an LLM alignment project for improving human-likeness in virtual patient dialogue and medical communication training.
- Prototyped DPO training components and supported data cleaning, synthetic dialogue generation, benchmarking, and debugging for SOPHIE-LLM.
- Built and debugged a lightweight agent framework connecting API-based and local LLM endpoints for batch synthetic data generation and structured output parsing.

### Goergen Institute for Data Science and AI, University of Rochester

*Undergraduate Research Assistant*

*Supervised by Prof. Cantay Caliskan*

**Rochester, NY**

*Jan 2025 – May 2025*

- Designed an end-to-end vegetation indexing pipeline integrating Mask2Former, MiDaS, and HSV-based refinement to classify trees, plants, and grass in complex urban scenes.

## PUBLICATIONS

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Hasan, M., **Zhao, J.** & Hoque, E. (2026). HAL: Inducing Human-likeness in LLMs with Alignment. arXiv Preprint. <https://arxiv.org/abs/2601.02813>

Caliskan, C., Chen, Z., Yang, L., Zhang, M., & **Zhao, J.** (2025). DEVISE: Depth-Aware Vegetation Indexing System. SSRN Working Paper. <https://ssrn.com/abstract=5371210> (Authors listed in alphabetical order.)

## FELLOWSHIPS AND GRANTS

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### Schwartz Discover Grant (\$5000)

Office of Undergraduate Research, University of Rochester 2025

- Project: Developing a Personality-aware Large Language Model for Human-Computer Interaction
- Mentor: Dr. Masum Hasan and Prof. Ehsan Hoque

### InnovateCS Research Grant (\$5000)

Department of Computer Science, University of Rochester 2026

- Project: An Open AR Platform for Human Centered Computer System Research
- Mentor: Prof. Yuhao Zhu

## TEACHING EXPERIENCE

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### Department of Computer Science, University of Rochester

Rochester, NY

Open Lab Support Teaching Assistant

Aug 2026 – Dec 2026

CSC 252 - Computer Organization Teaching Assistant

Jan 2026 – May 2026

CSC 242 - Introduction to Artificial Intelligence Teaching Assistant

Aug 2025 – Dec 2025

CSC 172 - Data Structures & Algorithms Summer Session Teaching Assistant

June 2025 – Aug 2025

CSC 172 - Data Structures & Algorithms Lab Teaching Assistant

Jan 2025 – May 2025

CSC 172 - Data Structures & Algorithms Workshop Leader

Aug 2024 – Dec 2024

## INTERSHIP EXPERIENCE

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### Institute of Automation, Chinese Academy of Sciences

Beijing, China

Artificial Intelligence Intern

May 2024 – Aug 2024

- Deployed the YOLOv5 object detection model on NVIDIA Jetson Nano board; Optimized performance for edge-based inference.
- Integrated the model with a monocular camera to accurately convey various road traffic signs, including directional arrows, traffic light, double amber lines, and so on.
- Designed and integrated control algorithms with the detection model to enable a smart car to automatically execute actions such as braking, steering, turning, and correcting its direction in real time.

## RESEARCH PRESENTATIONS

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### NY Architecture & Systems Day, Princeton University

Princeton, NJ

Lightning Talk & Poster Presenter

May 2026

- Presented “End-to-End Wireless Varifocal AR Prototype,” demonstrating world-locked rendering and multi-stream sensing on a wearable AR research platform.

## CAMPUS ACTIVITIES

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### Department of Computer Science, University of Rochester

Rochester, NY

Computer Science Undergraduate Council Tutor

Aug 2024 – May 2025

- Provided one-to-one and small-group tutoring in algorithms, data structures, and programming problem-solving.

## SKILLS

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- Programming Languages: Java, Python, C, HTML, JavaScript, Assembly
- Softwares: Photoshop, Premiere Pro, Autodesk Fusion, Zemax
- Machine Learning and AI Tools: Pytorch, Neural Network Model Development, Fine-Tuning, DPO, RLHF, SFT